

Juan Roberto García Gómez

Data Engineer | MES | Industrial IoT | Manufacturing Analytics

roberto.gag117@gmail.com
robertogag.github.io
LinkedIn
GitHub

PROFESSIONAL SUMMARY

Data Engineer with experience building manufacturing systems, MES, Industrial IoT, and operational analytics solutions. I develop production-ready software that connects shopfloor execution, structured data, and management decision-making. My work combines PostgreSQL, SQL, Node.js, React, ETL, Power BI, and manufacturing logic to deliver systems used daily by operations, supervisors, and management teams.

SELECTED IMPACT

- Built and deployed a web-based MES adopted for daily use across Injection, Assembly, EPC, and plant operations.
- Integrated 34 machines into real-time dashboards, historical analysis, and operational reporting workflows.
- Recovered MX\$15.2M in logistics claims within one year through structured analysis and evidence management.
- Delivered Power BI dashboards that reduced analysis time by 40% and improved management visibility.

PROFESSIONAL EXPERIENCE

Chunil Mexico

Oct 2024 - Present

Data Engineer | MES & Manufacturing Analytics

Led greenfield development of a manufacturing execution and analytics platform, replacing spreadsheet-based control with a centralized web application for daily plant use.

- Designed and delivered a production MES covering Injection, Assembly, EPC, Tooling, and plant execution workflows.
- Built PostgreSQL data structures and backend services for machines, parts, molds, shifts, cycle times, production targets, downtime, and quality data.
- Integrated 34 machines into real-time dashboards and historical workflows to support shopfloor response and management follow-up.
- Developed React dashboards and reporting logic for plan adherence, downtime, OEE, quality, and production control.
- Supported Industrial IoT integration and digital traceability across operational and reporting processes.

Ramírez Valle & Asociados | General Motors SLP Account

Oct 2023 - 2024

Claims, Loss & Recovery Coordinator

Managed operational analytics and financial recovery activities focused on claims analysis, supplier follow-up, evidence control, and executive reporting.

- Recovered MX\$15.2M in logistics claims within one year through structured case analysis and evidence management.
- Managed more than 100 investigations and created Power BI dashboards that reduced analysis time by 40%.
- Standardized documentation, tracking, and review routines to improve resolution time and reporting consistency.

CORE SKILLS

Data Engineering: PostgreSQL, SQL, ETL, data modeling, reporting logic, data quality.

Software Development: Node.js, Express, React, Vite, REST APIs, and operational web applications.

Manufacturing Systems: MES, ERP integration, Industrial IoT, OEE, downtime, plan adherence, quality reporting.

Analytics: Power BI, KPI design, dashboard development, operational performance analysis.

EDUCATION & PROFESSIONAL TRAINING

- **Full Stack Web Development Bootcamp, KeepCoding** | Nov 2025 - Present
- **Business Analytics, Tecnológico de Monterrey** | Jan 2025 - Apr 2025
- **Big Data, Artificial Intelligence & Machine Learning Bootcamp, KeepCoding** | Jun 2023 - Jun 2024
- **Diploma in Property and Casualty Claims, Instituto Mexicano Educativo de Seguros y Fianzas** | Feb 2024 - Nov 2024
- **Python for Data Science, IICO** | May 2022 - Aug 2022
- **Bachelor's Degree in Economics, Universidad Autónoma de San Luis Potosí** | Aug 2018 - Jul 2023
- **Bachelor's Degree in Industrial & Logistics Engineering, Universidad Interamericana para el Desarrollo** | Aug 2014 - Jul 2017

LANGUAGES

Spanish: Native

English: Professional working proficiency